

MASS-APP-014-V13 — Continuous Organizational Learning & Improvement Intelligence

Document Control

Field | Value

Document ID | MASS-APP-014-V13

Title | Continuous Organizational Learning & Improvement Intelligence

Version | 1.0

Status | Production Baseline v1.0

Authority | EWO-MASS-APP-014-V13

Application | MASS-APP-014 — Creative & Knowledge Intelligence

Date | 2026-08-03

1. Purpose

V13 transforms completed organizational work into governed, reusable learning that improves future human decisions and execution. Learning is tenant-scoped, evidence-backed, explainable, versioned, reviewable, and advisory. MASS may identify patterns and recommend improvement, but it never approves or executes organizational change.

2. Permanent Architecture

The permanent architecture provides five bounded capabilities:

1. Outcome Intake captures immutable references to completed work and verified outcomes.
2. Learning Analysis identifies successes, failures, recurring issues, opportunities, and candidate patterns.
3. Evidence Governance preserves lineage, confidence, contradiction, and version history.
4. Improvement Stewardship converts approved learning into recommendations and backlog candidates.
5. Executive Learning assembles cited improvement briefings and maturity observations.

V13 owns learning records, pattern hypotheses, evidence associations, recommendation review, maturity assessments, and improvement backlog proposals. It does not own source outcomes, enterprise knowledge persistence, analytical truth, operational workflows, policies, or execution.

3. Constitutional Responsibility

V13 supports Knowledge, Research, Enterprise Analytics, Executive Governance, Enterprise Automation, Enterprise Data Architecture, and Constitutional Evolution by preserving evidence-backed organizational understanding. It remains subordinate to constitutional ownership: departments own their responsibilities, executives retain authority, and historical truth cannot be rewritten by a learning process.

4. Platform Consumption Map

Source | Capability Consumed | Delegated Responsibility

APP-014 V01–V12 | Sessions, memory, decisions, agents, communication, execution, discovery, awareness, assurance | Prior volumes remain authoritative for their records

APP-013 | Projects, documents, components, templates, publications, assets, content, visualizations | Design Studio owns artifact stewardship

ENG-003 / ENG-004 | Principal and authorization decision | Identity and security remain platform-owned

ENG-005 | Learning lifecycle events | Event transport remains platform-owned

ENG-007 | Approved knowledge references | Knowledge stewardship remains ENG-007-owned

ENG-008 | Evidence and briefing document references | Persistence remains ENG-008-owned

ENG-009 | Governed analysis and advisory orchestration | AI orchestration remains ENG-009-owned

ENG-011 | Audit and observability | Platform telemetry remains ENG-011-owned

ENG-015 | API conventions | Interface governance remains ENG-015-owned

ENG-024 | Analytics, confidence inputs, trends | Enterprise analysis remains ENG-024-owned

ENG-027 | Lineage, ontology, information quality | Information governance remains ENG-027-owned

Events Published

? learning.outcome-captured

? learning.pattern-proposed

? learning.pattern-approved

? learning.recommendation-ready

? learning.recommendation-retired

? learning.contradiction-recorded

? learning.maturity-assessed

? learning.briefing-issued

Events Consumed

? Verified completion and assurance events from V12

? Decision outcome events from V03

? Workflow completion events from V09 and ENG-006

? Knowledge and lineage changes from ENG-007 and ENG-027

? Analytical observation events from ENG-024

Consumed events create observations; they never transfer source ownership.

5. Gateway Inventory

Gateway | Contract | Direction | Failure Behavior

App014ContextGateway | Read-only references to V01–V12 | Inbound | Mark source unavailable; do not fabricate context

DesignStudioArtifactGateway | APP-013 artifact metadata and versions | Inbound | Preserve reference and report stale/unavailable source

KnowledgeGateway | ENG-007 retrieval and knowledge contribution request | Bidirectional governed request | Queue contribution for review; never write directly

DocumentGateway | ENG-008 evidence and briefing persistence | Bidirectional governed request | Fail atomically and retain retry-safe job state

AIOrchestrationGateway | ENG-009 analysis/advisory request | Bidirectional request/response | Return evidence without synthesis when AI is unavailable

AnalyticsGateway | ENG-024 metrics and analytical observations | Inbound | Mark confidence incomplete and identify missing inputs

LineageGateway | ENG-027 lineage and quality validation | Inbound | Prevent approval when required lineage cannot be verified

EventGateway | ENG-005 publish/subscribe contract | Bidirectional | Use outbox and idempotent consumption

AuditGateway | ENG-011 audit and health contract | Outbound | Preserve local audit record and retry delivery

Gateways are adapters to authoritative contracts, not new platform services.

6. Role Mapping

V13 Role | APP-014 Baseline Role | Relationship | Permitted Authority

Learning Viewer | Viewer | Specializes | View authorized learning, recommendations, and briefings

Learning Contributor | Contributor | Specializes | Capture outcomes, propose lessons, and attach evidence

Learning Steward | Steward | Specializes | Validate evidence, review patterns, govern confidence, retire learning

Improvement Owner | Contributor | Extends | Accept ownership of an approved improvement backlog item

Executive Learning Approver | Administrator | Specializes | Approve recommendations and issue executive briefings

Learning Administrator | Administrator | Specializes | Configure tenant policies, gateways, and role assignments

These roles extend or specialize the baseline hierarchy; they do not replace it. Contributors cannot approve their own patterns or recommendations. AI, NOVA, POPS, agents, and plugins hold no approval role.

7. Organizational Learning Lifecycle

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Outcome Observed
→ Evidence Validated
→ Lesson Proposed
→ Pattern Evaluated
→ Human Review
→ Learning Approved
→ Recommendation Prepared
→ Human Decision
→ Improvement Backlog
→ Outcome Re-observed
→ Learning Refined or Retired
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Rejected hypotheses remain historically visible with their evidence and rationale. Contradictory evidence lowers confidence and reopens review; it never silently overwrites an approved record.

8. Outcome Capture

An outcome observation records tenant, source type, source identifier, source version, outcome type, completion time, verification state, responsible domain, and observed facts. Source content remains in its authoritative system. V13 stores only governed references, normalized observations, and lineage.

Outcomes must be verified before they support an approved learning record. Unverified outcomes may inform investigation but cannot raise confidence beyond the tenant policy threshold.

9. Lessons, Patterns, and Evidence

Lessons Learned

A lesson states what happened, why it matters, what evidence supports it, its scope, and its limitations. Lessons are hypotheses until reviewed.

Pattern Extraction

Patterns identify repeated relationships among conditions, actions, and outcomes. Each pattern records population, observation window, supporting and contradicting evidence, confidence method, confidence value, and applicability boundary.

Confidence

Confidence is an explainable assessment, not certainty. V1 uses configured evidence sufficiency, recurrence, source quality, contradiction, and recency weights. Confidence never authorizes execution.

Contradictory Evidence

Contradictions are first-class records linked to both the learning version and source evidence. Material contradictions change the learning state to reopened, block new recommendation approval, and require steward review.

10. Recommendations and Improvement Backlog

Recommendations translate approved learning into practical options. Each recommendation contains purpose, affected scope, evidence chain, expected benefit, risk, prerequisites, confidence, alternatives, and required human authority.

Approved recommendations may create backlog candidates. The backlog record is a proposal and does not create a workflow, task, policy, or operational change. Handoff to planning or execution occurs only after human approval through the owning application and platform capability.

11. Organizational Maturity

Maturity assessments summarize demonstrated practices across defined dimensions such as evidence quality, repeatability, outcome verification, exception recovery, knowledge reuse, and improvement follow-through. Scores retain indicator inputs and calculation versions. They cannot be used as undisclosed employment, relationship, or punitive determinations.

12. NOVA and POPS Interaction

NOVA improvement analysis and POPS historical context are advisory request types executed exclusively through ENG-009. NOVA may explain patterns and options. POPS may supply historical precedent and stewardship context. Their responses require citations and are stored as advisory evidence; neither may approve, retire, prioritize, or execute learning.

13. Data Model

Entity | Purpose | Immutable Condition

LearningProgram | Tenant learning policy and scope | Versioned after activation

OutcomeObservation | Normalized reference to completed work | Immutable after verification

EvidenceReference | Source evidence and lineage | Immutable after validation

LearningRecord | Governed lesson or learning identity | Identity preserved; content versioned

LearningVersion | Versioned learning statement | Immutable after approval/rejection

PatternHypothesis | Candidate recurring pattern | Immutable after review

PatternEvidence | Supporting or contradicting association | Immutable after review

ContradictionRecord | Material conflicting evidence | Immutable after resolution

ImprovementRecommendation | Evidence-backed improvement option | Immutable after decision

RecommendationReview | Human approval or rejection | Always immutable

ImprovementBacklogItem | Approved proposal for owning system | Immutable after handoff

MaturityAssessment | Point-in-time maturity observation | Immutable after issuance

ExecutiveImprovementBriefing | Cited executive learning summary | Immutable after issuance

AdvisoryRequest | NOVA or POPS request through ENG-009 | Response immutable after completion

All tenant-owned tables use DEFAULT gen_random_uuid(), UNIQUE(id, tenant_id), composite tenant-safe foreign keys, audit timestamps, and concrete auth.jwt()-based RLS.

14. API Inventory

Method | Path | Purpose | Minimum Role

POST | /learning/outcomes | Capture outcome observation | Learning Contributor

GET | /learning/outcomes | Search outcomes | Learning Viewer

POST | /learning/evidence | Register evidence reference | Learning Contributor

POST | /learning/records | Propose learning record | Learning Contributor

GET | /learning/records/{id} | Retrieve learning and versions | Learning Viewer

POST | /learning/records/{id}/versions | Propose revised learning | Learning Contributor

POST | /learning/versions/{id}/approve | Approve learning version | Learning Steward

POST | /learning/versions/{id}/reject | Reject learning version | Learning Steward

POST | /learning/patterns | Propose pattern | Learning Contributor

POST | /learning/patterns/{id}/review | Review pattern | Learning Steward

POST | /learning/contradictions | Record contradiction | Learning Contributor

POST | /learning/contradictions/{id}/resolve | Resolve contradiction | Learning Steward

POST | /improvement/recommendations | Prepare recommendation | Learning Steward

POST | /improvement/recommendations/{id}/decide | Approve or reject recommendation | Executive Learning Approver

POST | /improvement/recommendations/{id}/backlog | Create backlog candidate | Improvement Owner

GET | /improvement/backlog | Browse improvement candidates | Learning Viewer

POST | /learning/maturity-assessments | Create assessment | Learning Steward

GET | /learning/maturity-assessments/latest | Retrieve current assessment | Learning Viewer

POST | /learning/briefings | Prepare executive briefing | Learning Steward

POST | /learning/briefings/{id}/issue | Issue immutable briefing | Executive Learning Approver

Mutating requests require idempotency keys. Approval endpoints require distinct proposer and approver identities where separation of duty applies.

15. Security, Integrity, and Audit

? Tenant context derives from auth.jwt() -> app_metadata.tenant_id; client-supplied tenant identifiers are never trusted.

? Role claims derive from auth.jwt() -> app_metadata.roles and are checked by both API authorization and RLS write policies.

? Composite foreign keys prevent cross-tenant associations.

? Approved/rejected learning versions, reviewed patterns, resolved contradictions, recommendation decisions, handed-off backlog items, issued assessments, issued briefings, and completed advisory responses are database-immutable.

? Approval triggers prevent self-approval.

? Evidence source authorization is checked at registration and read time.

? Audit events include actor, tenant, correlation ID, prior state, new state, reason, and source references.

? Retirement ends active use but preserves institutional history.

Implementation-grade tables, constraints, indexes, triggers, and RLS policies are defined in MASS-APP-014-V13_Migration_Reference.sql.

16. Failure Behavior

? Missing source evidence leaves the observation unverified.

? Failed AI analysis returns the evidence package without inferred learning.

? Unavailable gateways produce explicit partial-analysis status.

? Conflicting evidence reopens learning review.

? Failed event publication is retained in the outbox.

? Duplicate requests return the prior idempotent result.

? Rejected downstream handoff remains a backlog candidate with failure evidence; V13 does not retry execution autonomously.

17. V1 Implementation

V1 implements tenant-scoped outcome capture, evidence references, versioned learning records, configurable confidence assessment, pattern review, contradiction handling, recommendation decisions, improvement backlog candidates, maturity snapshots, executive briefings, and NOVA/POPS advisory requests through ENG-009.

V1 uses explainable configured rules and human review. It does not implement autonomous model training, cross-tenant learning, automatic policy changes, or self-improving execution loops.

18. Future Evolution

Future versions may add richer longitudinal pattern analysis, expanded maturity models, additional governed evidence adapters, simulation-supported recommendation comparison, and privacy-preserving aggregate research. Each capability attaches through established gateway, evidence, version, and approval contracts. Future evolution cannot weaken tenant isolation, historical integrity, explainability, or human authority.

19. Constitutional Boundary Statement

V13 preserves and explains organizational learning. It does not own source truth, enterprise knowledge, analytics, policy, relationships, personnel decisions, operations, workflows, communication delivery, or executive authority. It may recommend, explain, prioritize, and preserve. It shall never approve, rewrite history, execute operational change, or create autonomous feedback loops.